MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, LOW POWER,
MONOSTABLE MULTIVIBRATOR, MONOLITHIC SILICON

This limited coordination Military specification has been prepared by the Air Force Space and Missile Systems Organization based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-M-38510/42(USAF). It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers the requirements for monolithic silicon, TTL, low power, monostable multivibrator microcircuits for use in Air Force missile and space systems.

1.2 Part number. The complete part number shall be as shown in the following example:

M38510	/042	01	S	T	D
Military designator	Detail specification	Device type (1.2.1)	Device class (1.2.2)	Case outline (1.2.3)	Lead finish (3.3)

1.2.1 Device type. The device type shall be as shown in the following:

Device type	Circuit	Similar commercial type
01	Single monostable multivibrator	54L121
02	Single retriggerable monostable multivibrator with clear input	54L122

1.2.2 Device class. The device class shall be Class S as defined in MIL-M-0038510.

1.2.3 Case outline. The case outline shall be as specified in figure 10 of MIL-M-0038510 and shall be identified as outline T.

1.3 Absolute maximum ratings.

Supply voltage range	0 Vdc to 8.0 Vdc
Input voltage range	0 Vdc to 5.5 Vdc
Storage temperature range	-65°C to 150°C
Lead temperature (soldering, 10 seconds)	300°C

1.4 Recommended operating conditions.

Ambient operating temperature range	-55°C to 125°C
Supply voltage	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage	2.0 Vdc
Maximum low level input voltage	0.8 Vdc

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2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-0038510 - Microcircuits, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-0038510 and as specified herein.

3.2 Design, construction, weight and physical dimensions. The design, construction, weight and physical dimensions shall be as specified in MIL-M-0038510 and herein.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

3.2.3 Schematic circuit. The schematic circuit shall be as specified on figure 3.

3.2.4 Case outline. The case outline shall be as specified in 1.2.3.

3.2.5 Weight. The maximum weight shall be 1 gram.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-0038510.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in tables I and II.

3.4.1 Delta drift limits. The delta drift limits shall be as specified in table I.

3.5 Screening. Screening shall be in accordance with table I of MIL-M-0038510 and as specified herein.

3.5.1 Interim electrical parameters. The interim electrical parameters shall consist of all the electrical characteristics tests as specified in table I. Each test shall be performed go, no-go at -55°C and 125°C. Each test shall also be read and recorded at 25°C. For all tests, the devices shall meet the limits specified in table I.

3.5.2 Final electrical tests. The final electrical tests shall consist of all the electrical characteristics tests specified in table I. Each test shall be read and recorded at -55°C, 25°C and 125°C. For all tests, the devices shall meet the limits specified in table I.

3.5.2.1 Delta computation. Delta computations shall be performed for those parameters in table I which are marked with an asterisk. All devices shall meet the applicable delta limits specified in table I. (See section 6 in MIL-M-0038510.)

3.5.3 Switching characteristics tests. The switching characteristics tests specified in table II and figures 4 and 5 shall be performed at 25°C. All devices shall meet the limits specified in table II.

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TABLE I. (Device type 01) Electrical characteristics tests.

1/ 2/ TEST NO.	TEST NAME	PIN NUMBERS														3/ LIMITS		1/ DELTA Δ	UNITS
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	MIN	MAX		
1*	I_{IH1}			2.4 V	GND	2.0 V		GND							5.5 V		20	10X or 2 µA	µA
2*	I_{IH1}			GND	2.4 V	2.0 V											20	10X or 2 µA	µA
3*	I_{IH2}			2.0 V	2.0 V	2.4 V											40	10X or 4 µA	µA
4	I_{IH3}			5.5 V	GND	2.0 V											1		mA
5	I_{IH3}			GND	5.5 V	2.0 V											1		mA
6	I_{IH3}			2.0 V	2.0 V	5.5 V											1		mA
7	I_{IL1}			0.4 V	2.0 V	GND											-.8		mA
8	I_{IL1}			2.0 V	0.4 V	GND											-.8		mA
9	I_{IL2}			GND	GND	0.4 V	A								5.5 V		-1.6		mA
10*	V_{OL}			0.4 V	0.4 V	0.4 V	8 mA								4.5 V		0.4	±0.04	V
11*	V_{OL}	8 mA		0.4 V	0.4 V	0.4 V				Pin 14	Y	GND					0.4	±0.04	V
12*	V_{OH}			0.4 V	0.4 V	0.4 V	-200 µA			Pin 14	Y	GND				2.4		±0.12	V
13*	V_{OH}	-200 µA		0.4 V	0.4 V	0.4 V									4.5 V	2.4		±0.12	V
14	I_{OS}	GND		GND	GND	GND									5.5 V	-10	-27		mA
15	I_{OS}			GND	GND	GND	GND			Pin 14	GND	GND				-10	-27		mA
16	I_{CC1}			GND	GND	GND				Pin 14							12		mA
17	I_{CC2}			GND	GND	GND				20K to Pin 14					5.5 V		20		mA
18	V_{IC}			-12 mA											4.5 V		-1.5		V
19	V_{IC}				-12 mA										4.5 V		-1.5		V
20	V_{IC}					-12 mA		GND							4.5 V		-1.5		V

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurement"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C, except that V_{IC} tests apply at T_A = 25°C only.

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TABLE I. (Device Type 02) Electrical characteristics tests.

Test No. 1/2/	Characteristics	Symbol	Conditions	Limits 3/ Min	Limits 3/ Max	Delta (Δ) 1/	Units
1	High level input voltage	V_{IH}		2			V
2	Low level input voltage	V_{IL}			0.8		V
3	Input clamp voltage 8/	V_{IC}	V_{CC} = 4.5 V, I_{IC} = -12 mA		-1.5		V
4*	High level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -400 µA 5/	2.4		±0.12	V
5*	Low level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 8 mA 5/		0.4	±0.04	V
6	Input current at maximum input voltage	I_I	V_{CC} = 5.5 V, V_{IN} = 5.5 V		1		mA
7*	High level input current — Data inputs	I_{IH}	V_{CC} = 5.5 V, V_{IN} = 2.4 V		20	10X or 4 µA	µA
	High level input current — $\overline{\text{Clear}}$ inputs				40	10X or 4 µA	µA
8	Low level input — Data inputs	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V		-0.8		mA
	Low level input — $\overline{\text{Clear}}$ inputs				-1.6		mA
9	Short circuit output current 4/	I_{OS}	V_{CC} = 5.5 V 5/	-5.0	-20		mA
10	Supply current (quiescent or triggered)	I_{CC}	V_{CC} = 5.5 V 6/ 7/		14		mA

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurement"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.
4/ Not more than one output should be shorted at a time.
5/ Ground C_{ext} to measure V_{OH} at Q, V_{OL} at $\overline{Q}$, or I_{OS} at Q. C_{ext} is open to measure V_{OH} at $\overline{Q}$, V_{OL} at Q, or I_{OS} at $\overline{Q}$.
6/ Quiescent I_{CC} is measured (after clearing) with 2.4 V applied to all clear and A inputs, B inputs grounded, all outputs open, C_{ext} = 0.02 µF, and R_{ext} = 25 kΩ. R_{int} is open.
7/ I_{CC} is measured in the triggered state with 2.4 V applied to all clear and B inputs, A inputs grounded, all outputs open, C_{ext} = 0.02 µF, and R_{ext} = 25 kΩ. R_{int} is open.
8/ T_A = 25°C only.

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TABLE II. (Device Type 01) Switching characteristics.

Characteristic	Symbol	Conditions	Test Figure	Limits		Units
				Min	Max	
Propagation delay time to logical 1 level from B input to Q output	t_{PLH}	C_L = 15 pF, C_T = 80 pF	4	---	110	ns
Propagation delay time to logical 1 level from A1/A2 inputs to Q output	t_{PLH}			---	140	ns
Propagation delay time to logical 0 level from B input to Q output	t_{PHL}	C_L = 15 pF, C_T = 80 pF	4	---	130	ns
Propagation delay time to logical 0 level from A1/A2 inputs to Q output	t_{PHL}			---	160	ns
Pulse width obtained using internal timing resistor	$t_{p(out)}$	C_L = 15 pF, C_T = 80 pF, R_T = open, Pin 9 to V_{CC}	5	70	260	ns
Pulse width obtained with zero timing capacitance	$t_{p(out)}$	C_L = 15 pF, C_T = 0, R_T = open, Pin 9 to V_{CC}	5	20	70	ns
Pulse width obtained using external timing resistor	$t_{p(out)}$	C_L = 15 pF, C_T = 100 pF, R_T = 10 kΩ, Pin 9 open	5	600	850	ns
		C_L = 15 pF, C_T = 1 µF, R_T = 10 kΩ, Pin 9 open	5	6	8	ms
Minimum recommended operating duration of trigger pulse	$t_{(hold)}$	C_L = 15 pF, C_T = 80 pF, R_T = open, Pin 9 to V_{CC}	5	---	100	ns

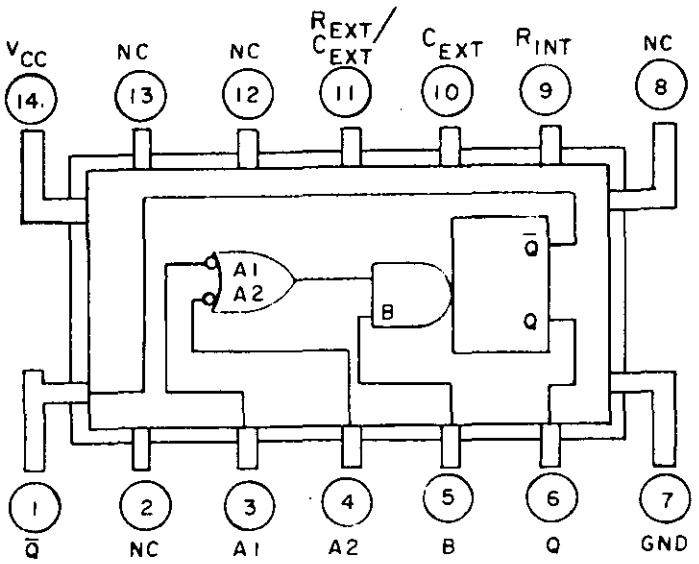
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TABLE II. (Device Type 02) Switching characteristics.

Characteristic	Symbol	Conditions	Test Figure	Limits		Units
				Min	Max	
Propagation delay time, low-to-high-level Q output, from either A input	t_{PLH}	C_{ext} = 0, R_{ext} = 5 kΩ, C_L = 15 pF, R_L = 800 ohms, V_{CC} = 5 V, T_A = 25°C	4	---	66	ns
Propagation delay time, low-to-high-level Q output, from either B input	t_{PLH}		4	---	56	ns
Propagation delay time, high-to-low-level Q output, from either A input	t_{PHL}		4	---	80	ns
Propagation delay time, high-to-low-level Q output, from either B input	t_{PHL}		4	---	72	ns
Propagation delay time, high-to-low-level Q output, from clear input	t_{PHL}		4	---	54	ns
Propagation delay time, low-to-high-level Q output, from clear input	t_{PLH}		4	---	80	ns
Minimum width of Q output pulse	$t_{w(min)}$		4	---	130	ns
Width of Q output pulse	t_w	C_{ext} = 400 pF, R_{ext} = 10 kΩ, C_L = 15 pF, R_L = 800 ohms, V_{CC} = 5 V, T_A = 25°C	4	1.7	2.1	µs

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Device type 01

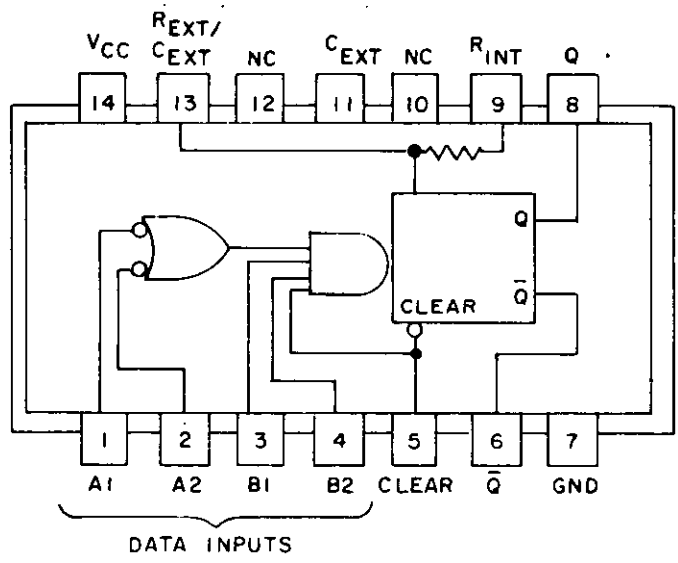


NOTES:
1. To use internal timing resistor (20 kΩ), connect R_{INT} to V_{cc}.
2. An external timing capacitor may be connected between C_{EXT} and R_{EXT}/C_{EXT} (positive).
3. For accurate repeatable pulse widths, connect and external resistor between R_{EXT}/C_{EXT} and V_{cc}, with R_{INT} open-circuited.
4. To obtain variable pulse width connect variable resistance between R_{INT} or R_{EXT}/C_{EXT} and V_{cc}.

FIGURE 1. Logic diagrams and terminal connections.

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Device type 02



NOTES:
1. To use internal timing resistor (20 kΩ), connect R_{INT} to V_{cc}.
2. An external timing capacitor may be connected between C_{EXT} and R_{EXT}/C_{EXT} (positive).
3. For accurate repeatable pulse widths, connect and external resistor between R_{EXT}/C_{EXT} and V_{cc}, with R_{INT} open-circuited.
4. To obtain variable pulse width connect variable resistance between R_{INT} or R_{EXT}/C_{EXT} and V_{cc}.

FIGURE 1. Logic diagrams and terminal connections - Continued.

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Device type 01

t_n INPUT			t_{n+1} INPUT			OUTPUT
A_1	A_2	B	A_1	A_2	B	
H	H	L	H	H	H	INHIBIT
L	X	H	L	X	L	INHIBIT
X	L	H	X	L	L	INHIBIT
L	X	L	L	X	H	ONE SHOT
X	L	L	X	L	H	ONE SHOT
H	H	H	X	L	H	ONE SHOT
H	H	H	L	X	H	ONE SHOT
X	L	L	X	H	L	INHIBIT
L	X	L	H	X	L	INHIBIT
X	L	H	H	H	H	INHIBIT
L	X	H	H	H	H	INHIBIT
H	H	L	X	L	L	INHIBIT
H	H	L	L	X	L	INHIBIT

NOTES:
1. t_n = time before input transition.
2. t_{n+1} = time after input transition.
3. X indicates that either a low or high level may be present.
4. A1 and A2 are negative-edge-triggered logic inputs, and will trigger the one shot when either or both go to low level with B at high level.
5. B is A positive Schmitt-Trigger input for slow edges or level detection, and will trigger the one shot when B goes to logical H with either A1 or A2 at low level.

Device type 02

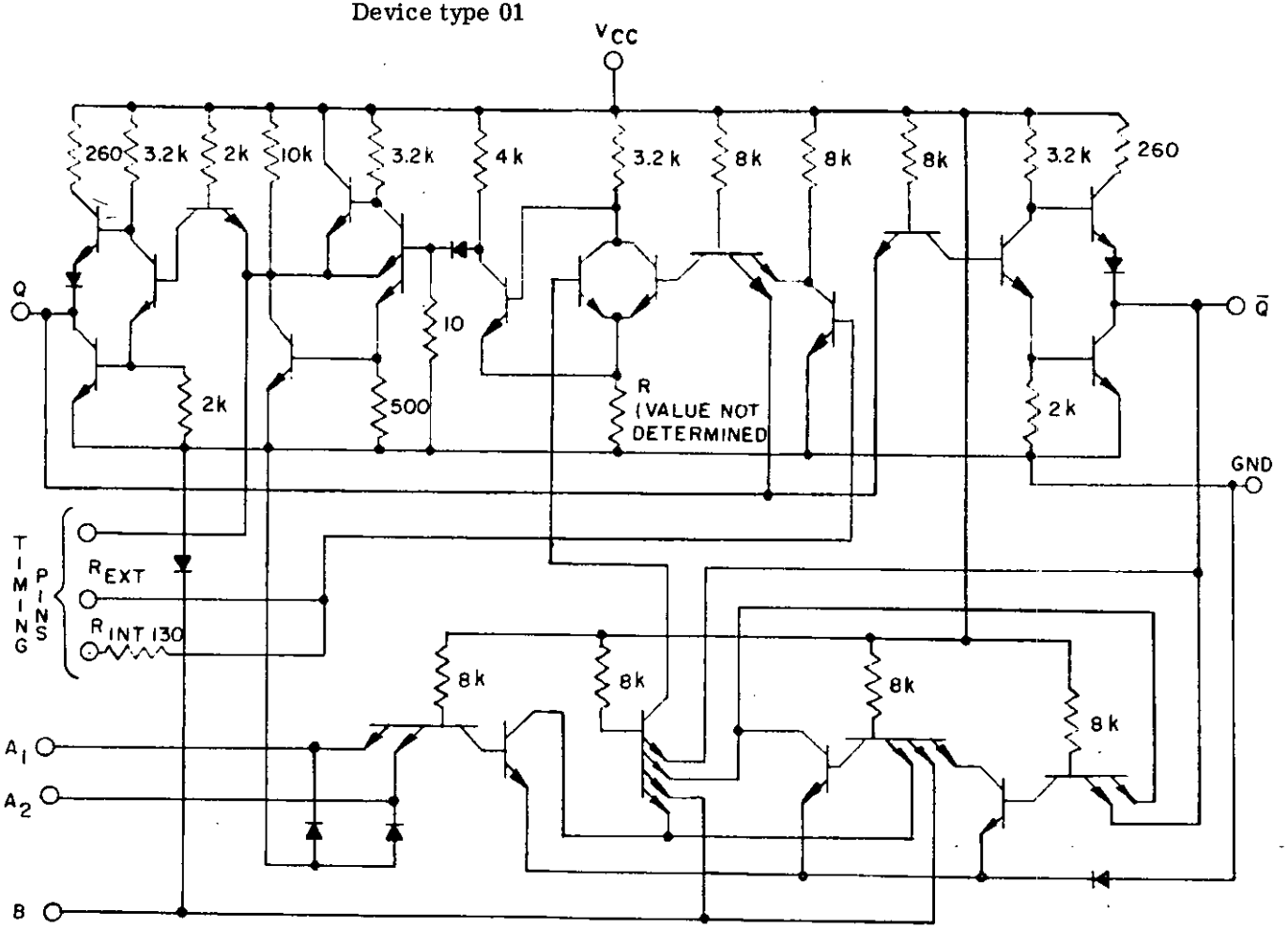
INPUTS				OUTPUTS	
A1	B2	B1	B2	Q	$\bar{Q}$
H	H	X	X	L	H
X	X	L	X	L	H
X	X	X	L	L	H
L	X	H	H	L	H
L	X	↓	H	⊓	⊔
L	X	H	↓	⊓	⊔
X	L	H	H	L	H
X	L	↓	H	⊓	⊔
X	L	H	↓	⊓	⊔
H	↓	H	H	⊓	⊔
↓	↓	H	H	⊓	⊔
↓	H	H	H	⊓	⊔

NOTES:
H = high level (steady state), L = low level (steady state), ↓ = transition from low to high level, ↓ = transition from high to low level, Ω = one high-level pulse. ⊓ = one low-level pulse, x = irrelevant (any input, including transitions).

FIGURE 2. Truth tables.

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Device type 01



NOTE: All resistor values nominal and expressed in ohms.

FIGURE 3. Schematic circuit.

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Device type 02

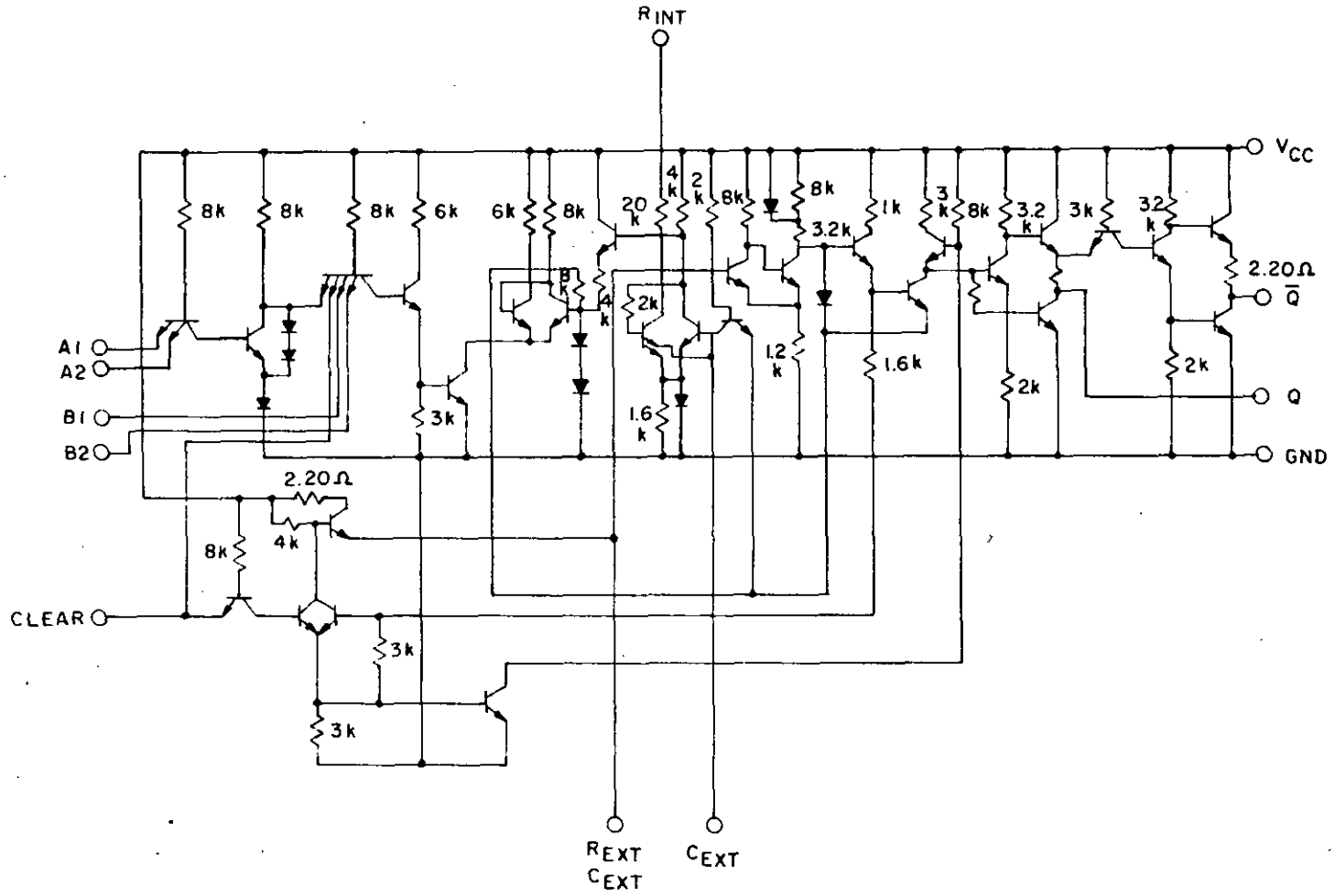
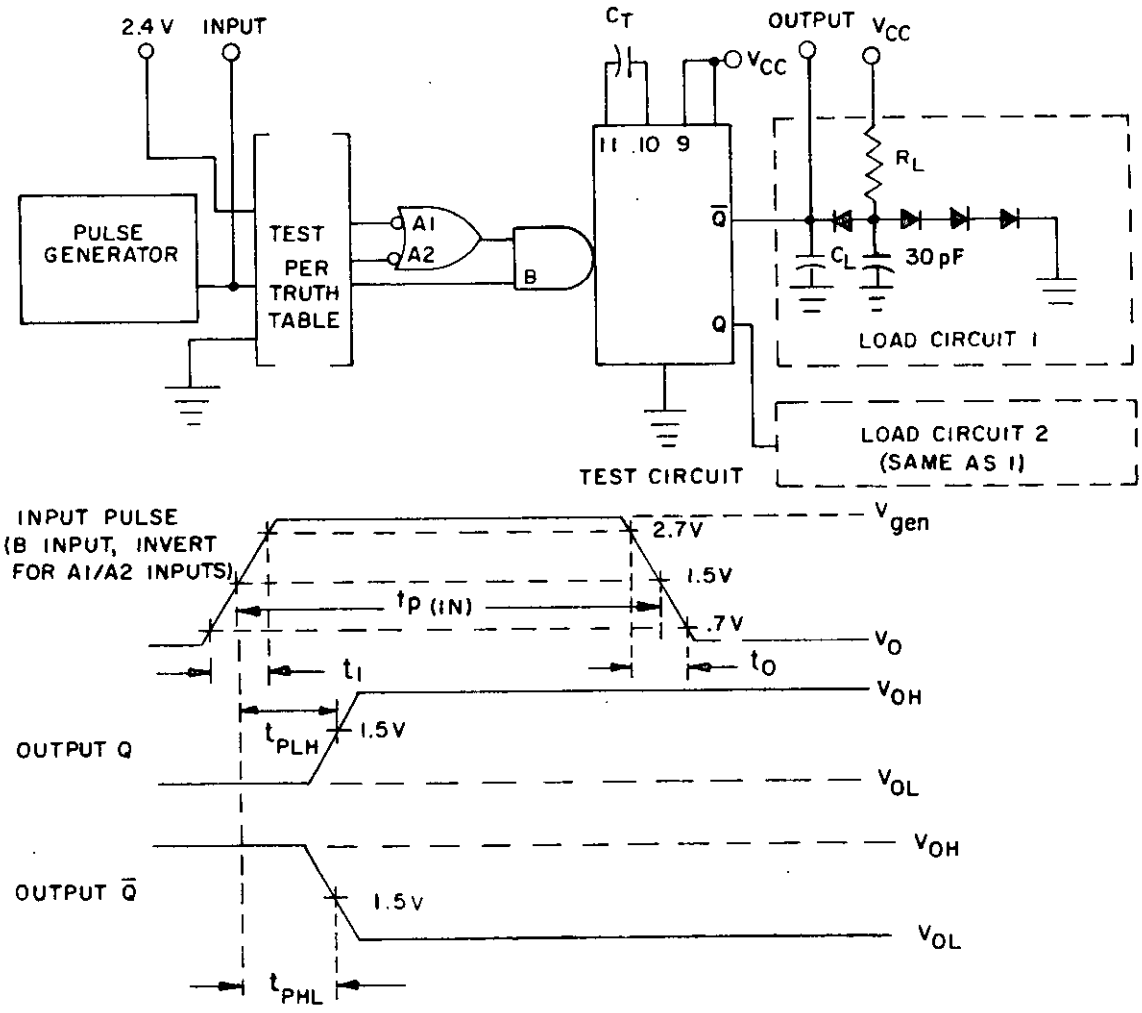


FIGURE 3. Schematic circuit - Continued.

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Device type 01

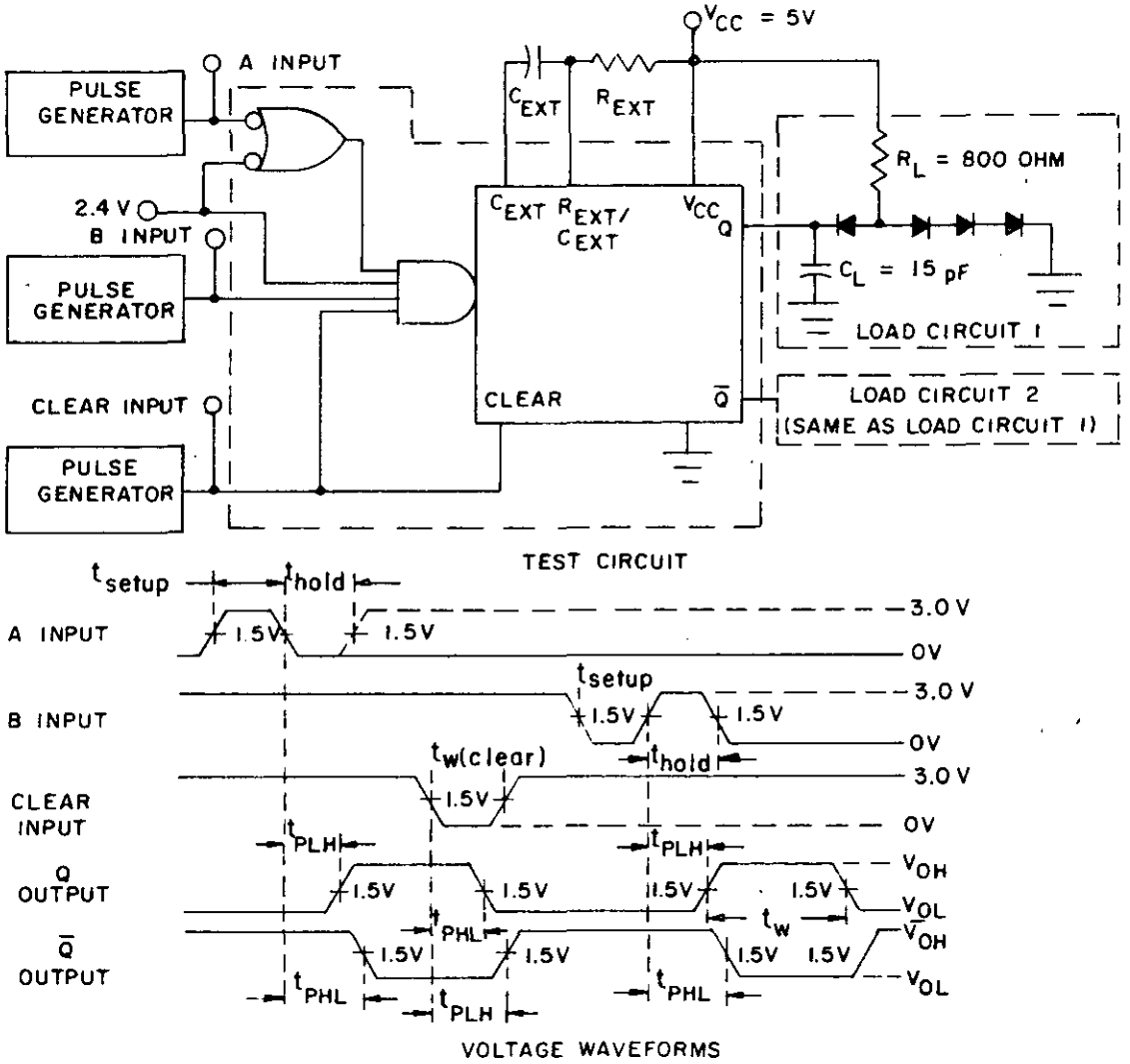


NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3.0 V, $t_0 \leq 10$ ns, t_1 = 10 ns, $t_{p(IN)} \geq 100$ ns, PRR = 1 MHz, and $Z_{OUT} \approx 50\ \Omega$.
2. All diodes are 1N3064, or equivalent.
3. C_L = 15 pF minimum including scope probe, wiring, and stray capacitance, without package in test fixture.
4. R_L = 800 Ω ±5 percent.

FIGURE 4. Switching test circuit and waveforms.

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Device type 02



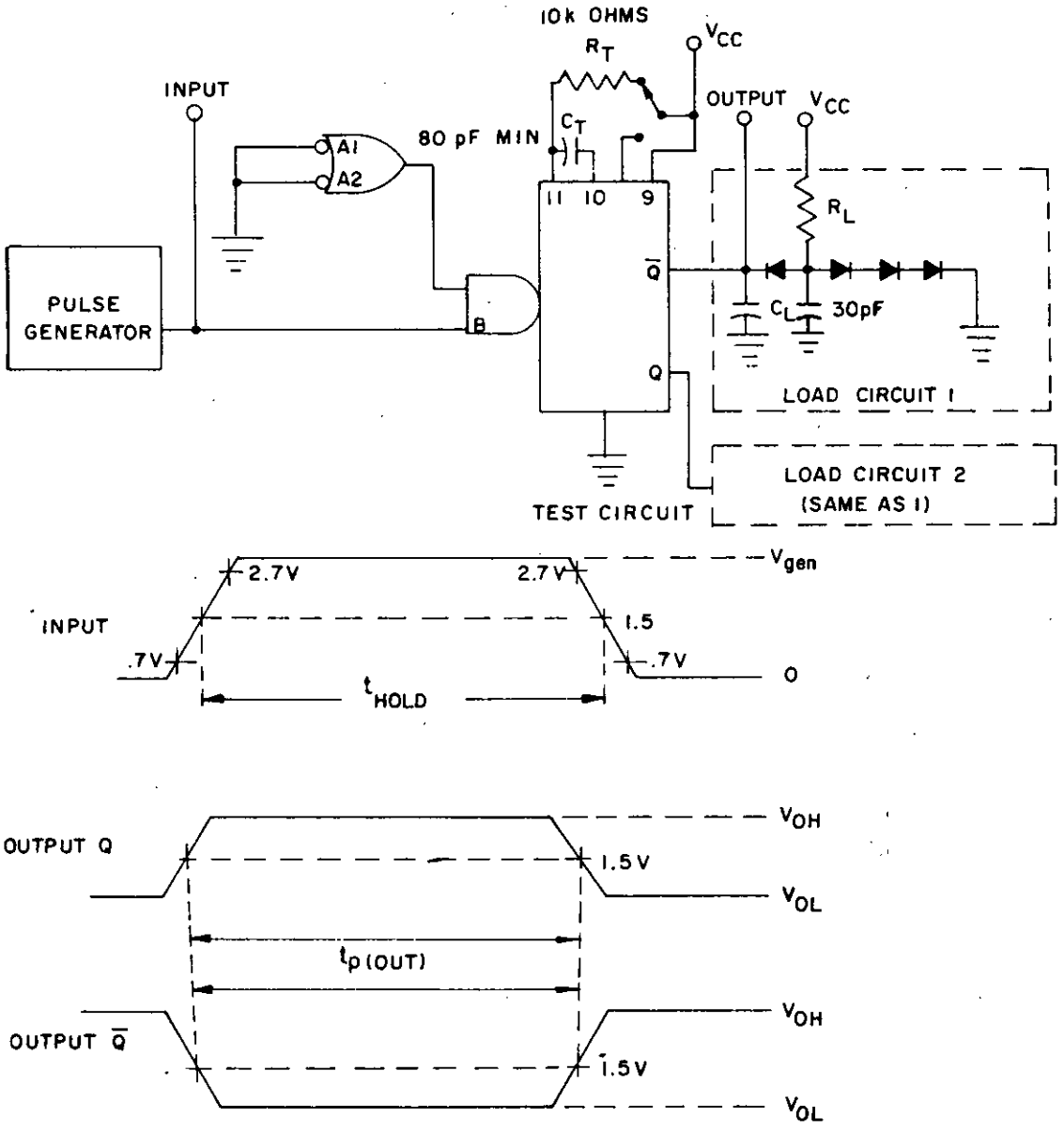
NOTES:

1. The pulse generators have the following characteristics: $t_r \leq 10$ ns (10 percent to 90 percent level), $t_f \leq 10$ ns, PRR ≤ 1 MHz, duty cycle ≤ 50 percent, $Z_{OUT} \approx 50$ ohms.
2. See test conditions, switching characteristics table II, for values of R_{ext} and C_{ext}.
3. All diodes are 1N916.
4. C_L = 15 pF minimum including scope probe, wiring, and stray capacitance, without package in test fixture.

FIGURE 4. Switching test circuit and waveforms - Continued.

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Device type 01



NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3.0 V, $t_0 \leq 10$ ns, $t_1 \leq 10$ ns, t_p = 50 ns, PRR = 1 MHz, and $Z_{OUT} \approx 50$ ohms.
2. All diodes are 1N3064, or equivalent.
3. C_L = 15 pF minimum including scope probe, wiring, and stray capacitance, without package in test fixture.
4. R_L = 400 ohms ±5 percent.

FIGURE 5. Test Circuit for t_{HOLD} and $t_{p(OUT)}$.

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3.6 Extended life tests. Extended life tests shall be performed in accordance with MIL-M-0038510.

3.7 Marking. Marking shall be in accordance with MIL-M-0038510 and 1.2 herein.

4. PRODUCT ASSURANCE PROVISIONS

4.1 Product assurance provisions. Product assurance shall be in accordance with MIL-M-0038510.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be as specified in MIL-M-0038510.

6. NOTES

6.1 Notes. Notes are as specified in MIL-M-0038510.

Custodian:
Air Force - 19

Review activities:
Air Force - 17, 19
DSA - ES

Preparing activity:
Air Force - 19

Agent:
DSA - ES

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